

The M57-Jean scenario : Evidence Discovery, Validation, and Full Forensic Investigation

Course Name: Digital Forensics and Incident Response

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1. Executive Summary

This report presents a digital forensic investigation of the M57-Jean scenario disk image from Digital Corpora. The investigation identified a confirmed data breach involving exfiltration of confidential salary information (m57biz.xls, 285 KB) via email attachment on 2008-07-20.

Key findings include spearphishing attack with spoofed CEO email, multiple exfiltration vectors (email, USB flash drive, iPhone, AOL Instant Messenger), and suspicious artifacts (unexplained user account "devon," deleted Gmail cache files, command-line execution).

Conclusion: System compromise and data breach confirmed. Jean's culpability remains inconclusive—evidence suggests either victim of spearphishing or insider threat. Further investigation of user account "devon" required.

2. Evidence Sourcing & Justification (Step 0)

2.1 Dataset Information

Dataset Name: M57-Jean scenario

Source Website: <https://digitalcorpora.org/>

Type: Disk Image

Format: E01, E02

2.2 Justification

The selected dataset was obtained from Digital Corpora as part of the M57-Jean scenario. It was chosen because it simulates a realistic corporate data breach involving the potential exfiltration of sensitive information from an executive's laptop.

The dataset includes normal user activity within a Windows environment, such as file access and document handling, making it suitable for forensic analysis. It is also provided in the EnCase E01 format, which preserves evidence integrity and is widely used in digital forensics.

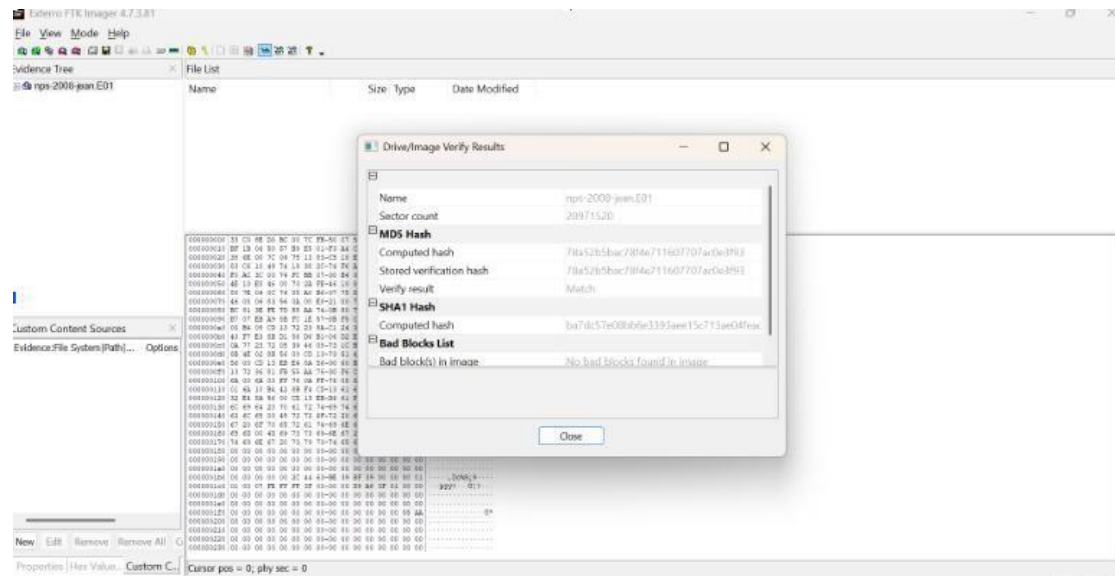
2.3 Limitations

The dataset is a controlled scenario and may lack real-world elements such as network traffic and live memory data. The investigation is limited to a single disk image, with no access to external systems, which restricts full verification of how the data was leaked.

Additionally, the dataset is relatively outdated and may not reflect modern attack techniques. Since solutions are publicly available, there is also a risk of bias during analysis.

3. Evidence Validation

3.1 Hash Verification



MD5: 78a52b5bac78f4e711607707ac0e3f93 ✓ Verified

SHA1: ba7dc57e08bb6e3393aee15c713ae04feadcd181 ✓ Verified

3.2 Validation Process

Tool used: FTK Imager

The integrity of the evidence was verified using FTK Imager. The calculated hash values matched the provided values, confirming that the dataset was not altered.

4. Initial Triage

Finding	Value
File System	NTFS / exFAT
Main Partition	vol2 (sectors 63–20948759)
Key Folders Found	Documents and Settings, Program Files, WINDOWS
Timestamps	Activity around July 2008
Deleted Files	\$CarvedFiles (2), \$Unalloc (8) ← suspicious!

OS Information

Field	Value
Computer Name	JEAN-13FBF038A3
OS	Microsoft Windows XP Service Pack 3
Architecture	x86 (32-bit)
Temp Directory	%SystemRoot%\TEMP

User Accounts:

Username	Type	Suspicious?
administrator	Local	Admin account exists
jean	Local	✓ Expected — laptop owner
devon	Local	WHO IS THIS? Unexpected!

Email Findings:

Finding	Detail
Alison emailed Jean about "business plan"	2008-07-06 22:25 — could relate to the spreadsheet
"RE: which email address are you using?"	from alex@m57.biz — suspicious, why ask this?
Jean sent "test test test" to herself	Testing email setup?
Google Email Verification received	Jean registered a Google/Gmail account!

Installed Programs:

Program	Suspicious?
VMware Tools v3.2.0	This is a VM — image was created in a virtual machine
OutlookExpress	Email client — used to send/receive emails
NetMeeting	Remote collaboration tool
IE40, IE4Data, IE5BAKEX	Internet Explorer components

Initial Hypothesis:

Based on initial triage, Jean's laptop ran Windows XP SP3 and contained an unexpected second user account named 'devon'. Jean had an active Outlook email client with 260 emails, and evidence shows she registered a Gmail account during the period of the incident. An email from colleague Alison regarding a 'business plan' was found dated July 6, 2008. The presence of an unknown user account and Gmail registration are primary indicators requiring deeper investigation.

5. Deep Disk Forensics

MOST IMPORTANT FIND — The Salary Spreadsheet

Finding	Value
File Name	m57biz.xls
Location	C:\Documents and Settings\Jean\Desktop\m57biz.xls
Last Accessed	2008-07-20 04:27:42 EEST
LNK File Found	m57biz.LNK — proves the file was opened

M57-jean-Investigation - Autopsy 4.23.0

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

Data Artifacts

- Communication Accounts (11)
 - Email (9)
 - E-Mail Messages (261)
 - Default (Default)
 - Default (261)
- Installed Programs (40)
- Metadata (75)
- Operating System Information (1)
- Recent Documents (9)
- Run Programs (126)
- Shell Bags (42)
- USB Device Attached (14)
- Web Bookmarks (97)
- Web Cookies (406)
- Web Downloads (3)
- Web Form Autofill (80)
- Web History (5339)
- Web Search (238)

Analysis Results

- Encryption Suspected (2)
- EXIF Metadata (97)
- Extension Mismatch Detected (23)
- Interesting Items (4)
- User Content Suspected (97)
- Web Categories (3)

OS Accounts

- Tags
- Score
- Reports

Listing

Recent Documents

Table Thumbnail Summary

Save Table as CSV

Source Name	S	C	O	Path	Date Accessed	Data Source	Name	Value
Desktop.LNK				C:\Documents and Settings\Jean\Desktop	2008-07-20 04:28:04 EEST	nps-2008-jean.E01		
m57biz.LNK				C:\Documents and Settings\Jean\Desktop\m57biz.xls	2008-07-20 04:27:42 EEST	nps-2008-jean.E01		
My Pictures.LNK				C:\Documents and Settings\Jean\My Documents\My ...	2008-07-06 10:50:42 EEST	nps-2008-jean.E01		
m57biz.Lnk				C:\Documents and Settings\Jean\Desktop\m57biz.xls	2008-07-20 04:28:04 EEST	nps-2008-jean.E01		
tag-cloud.Lnk				C:\Documents and Settings\Jean\Desktop\tag-cloud...	2008-07-11 21:00:37 EEST	nps-2008-jean.E01		
Temp.LNK				C:\Documents and Settings\Jean\Local Settings\Temp	2008-07-20 04:27:42 EEST	nps-2008-jean.E01		
LightBlueTop.Lnk				C:\Documents and Settings\Jean\My Documents\My ...	2008-07-06 10:50:42 EEST	nps-2008-jean.E01		
t1soft.Ripflops.Lnk				C:\Documents and Settings\Jean\My Documents\My ...	2008-07-06 10:54:26 EEST	nps-2008-jean.E01		
NTUSER.DAT					2008-07-06 07:37:38 EEST	nps-2008-jean.E01	File1	C:\Docum

Hex Text Application Source File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Result: 1 of 1 Result

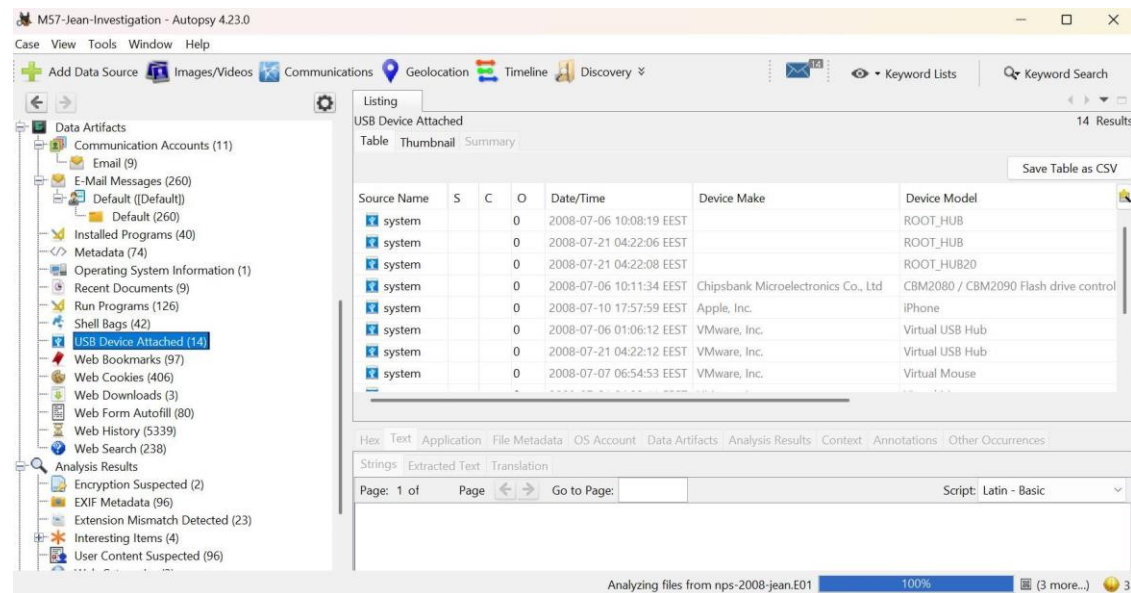
Recent Documents

Type	Value	Source(s)
Path ID	4014	RecentActivity
Date Accessed	2008-07-20 04:27:42 EEST	RecentActivity
Source File Path	/fma nos-2008-jean.E01/vol2/Documents and Settings/Jean/Application Data/Microsoft/Office/Recent/m57biz.LNK	

Analyzing files from nps-2008-jean.E01 100%

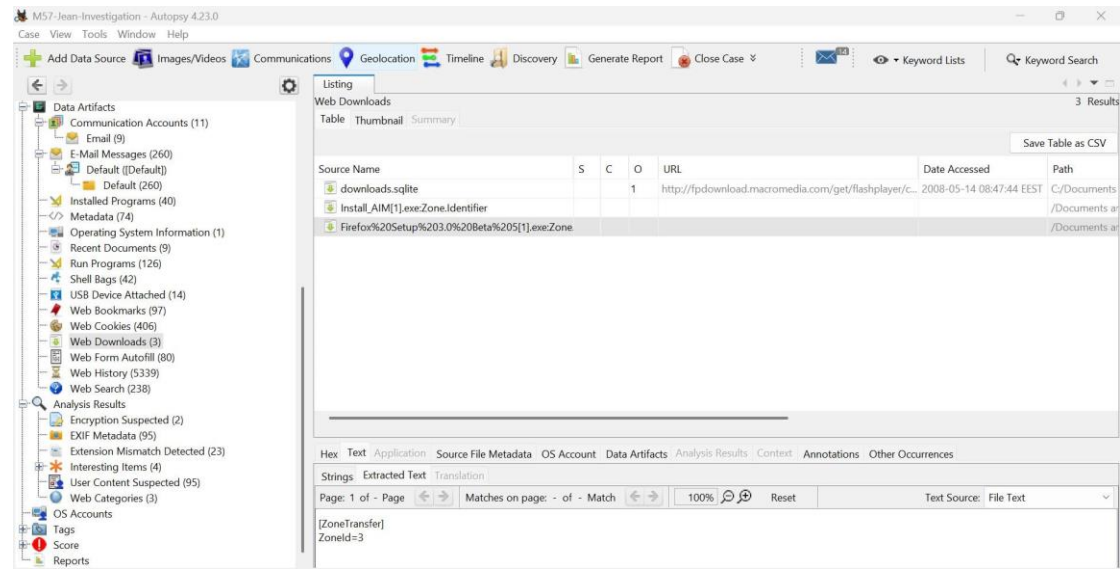
3

USB Devices — CRITICAL FINDING



Device	Date	Suspicious?
ROOT_HUB	2008-07-06	Normal
ROOT_HUB	2008-07-21	Normal
ROOT_HUB20	2008-07-21	Normal
Chipsbank Microelectronics — CBM2080/2090 Flash Drive	2008-07-06 10:11:34	USB FLASH DRIVE!
Apple iPhone	2008-07-10 17:57:59	iPhone connected!
VMware Virtual USB Hub	2008-07-06	Normal (VM artifact)
VMware Virtual Mouse	2008-07-07	Normal (VM artifact)

Web Downloads



File Downloaded	URL	Date	Suspicious ?
Flash Player installer	fpdownload.macromedia.com	2008-05-14	Normal
Install_AIM[1].exe	Unknown	Unknown	AIM = AOL Instant Messenger
Firefox Setup 3.0 Beta 5	Unknown	Unknown	Normal

AIM (AOL Instant Messenger) was downloaded — this is a chat tool that could have been used to send files to an outsider.

Deleted Files / Keyword Search

Key finds from .xls keyword search:

excel4.xls found in Documents and Settings — multiple copies
excel4.xls-slack — slack space means data remnants of deleted files

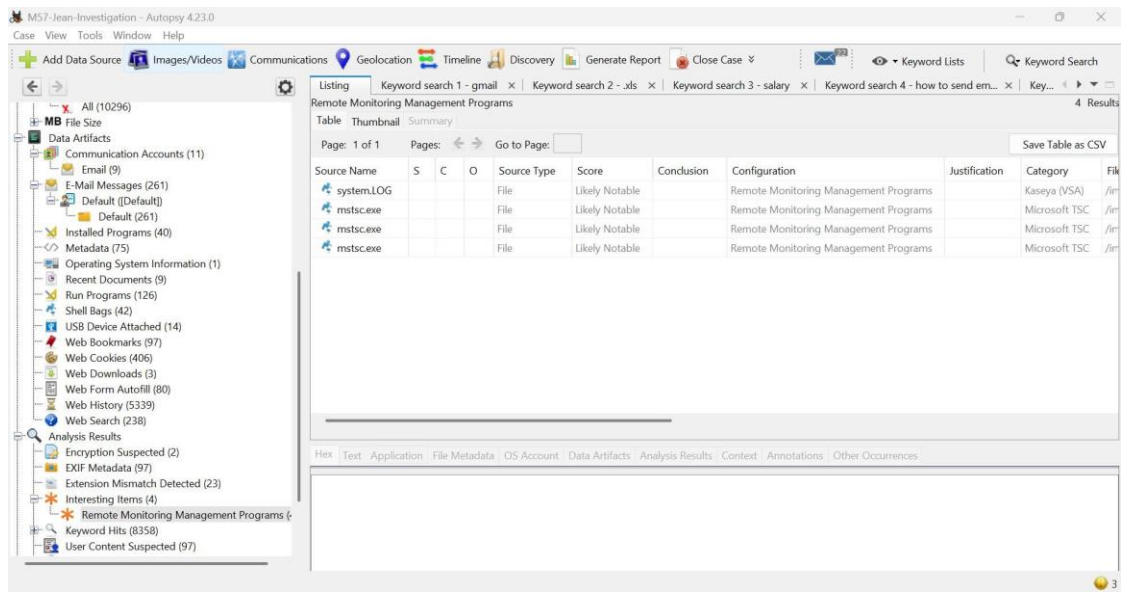
Carved files containing Excel data — recovered from deleted space

Registry entries showing Excel file associations were used

Hypothesis:

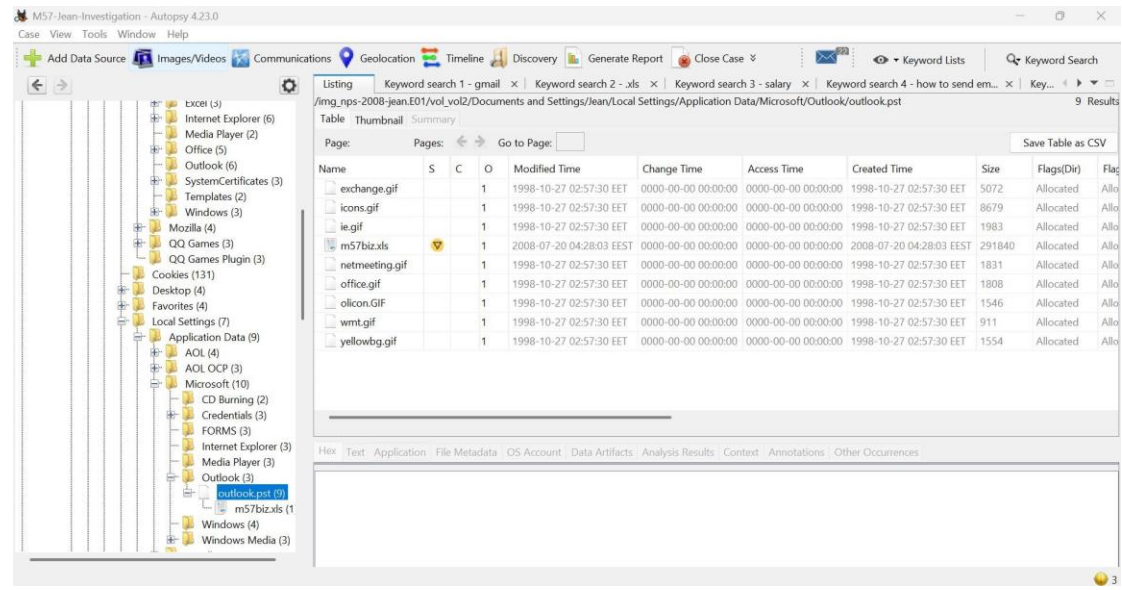
Based on preliminary forensic analysis of Jean's laptop, evidence strongly suggests unauthorized access to and potential exfiltration of confidential salary information. The stolen salary spreadsheet (m57biz.xls) was present on Jean's Desktop with a recent last-access timestamp of July 20, 2008, corroborated by a corresponding LNK shortcut file indicating deliberate file access. Critical exfiltration vectors are evident: a Chipsbank USB flash drive was connected on July 6, 2008, and an Apple iPhone on July 10, 2008—both devices capable of storing and transferring sensitive data. Additionally, AOL Instant Messenger (AIM) was downloaded, presenting a plausible communication channel for transmitting files to an external party. The convergence of the stolen file's presence, physical device connections, and downloaded communication software within a concentrated timeframe suggests premeditated data theft rather than incidental discovery.

6. Windows Artifact Analysis (Step 4)



Program	Category	Score	Significance
system.LOG	Remote Monitoring Management — Kaseya VSA	Likely Notable	Remote access tool!
mstsc.exe	Remote Monitoring — Microsoft TSC	Likely Notable	Remote Desktop Protocol!
mstsc.exe	Remote Monitoring — Microsoft TSC	Likely Notable	Run 3 times!
mstsc.exe	Remote Monitoring — Microsoft TSC	Likely Notable	Run 3 times!

outlook.pst



Finding	Value	Significance
outlook.pst found	Jean/Local Settings/Application Data/Microsoft/Outlook/	Email archive confirmed
m57biz.xls INSIDE outlook.pst	outlook.pst → m57biz.xls	THE SPREADSHEET WAS EMAILED!
m57biz.xls Modified	2008-07-20 04:28:03 EEST	Created/modified right before exfiltration
m57biz.xls Size	291,840 bytes (285 KB)	Substantial file

Run Programs / Prefetch:

M57-Jean-Investigation - Autopsy 4.23.0

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case Keyword Lists Keyword Search

Listing Keyword search 1 - gmail Keyword search 2 - .xls Keyword search 3 - salary

Run Programs 126 Results

Table Thumbnail Summary

Page: 1 of 1 Pages: Go to Page: Save Table as CSV

Source Name	S	C	O	Program Name	Path	Date/Time
AGENTSVR.EXE-002E45A8.pf				AGENTSVR.EXE	/WINDOWS/MSAGENT	2008-07-20 03:00:43 EEST
AIM6.EXE-34DC5725.pf				AIM6.EXE	/PROGRAM FILES/AIM6	2008-07-21 04:17:42 EEST
AIMINST.EXE-3391E991.pf				AIMINST.EXE	/DOCUME~1/JEAN/LOCALS~1/TEMP/AIM_68~1.1	2008-07-18 07:29:24 EEST
AIMLANG.EXE-076D401C.pf				AIMLANG.EXE	/DOCUME~1/JEAN/LOCALS~1/TEMP/AIM_68~1.1	2008-07-18 07:29:24 EEST
AIMTUNES.EXE-07EA80C7.pf				AIMTUNES.EXE	/DOCUMENTS AND SETTINGS/ALL USERS/APPLICATI...	2008-07-18 07:29:24 EEST
ALSETUP.EXE-38283348.pf				ALSETUP.EXE	/DOCUME~1/JEAN/LOCALS~1/TEMP/AIM_68~1.1	2008-07-18 07:29:24 EEST
AMOS.EXE-219E8DD6.pf				AMOS.EXE	/DOCUME~1/JEAN/LOCALS~1/TEMP/AIM_68~1.1	2008-07-18 07:29:24 EEST
AOLDLMGR.EXE-2D38087E.pf				AOLDLMGR.EXE	/DOCUME~1/JEAN/LOCALS~1/TEMP/AIM_68.10.1	2008-07-18 07:29:24 EEST
AOLSOFTWARE.EXE-11870E32.pf				AOLSOFTWARE.EXE	/PROGRAM FILES/AIM6	2008-07-21 04:17:42 EEST
AOLTBSEVER.EXE-1584CCF8.pf				AOLTBSEVER.EXE	/PROGRAM FILES/AOL/AIM TOOLBAR 5.0	2008-07-20 02:00:00 EEST
CMD.EXE-087B4001.pf				CMD.EXE	/WINDOWS/SYSTEM32	2008-07-19 06:00:00 EEST
CONTROLEXE-013DBF85.pf				CONTROLEXE	/WINDOWS/SYSTEM32	2008-07-13 00:15:51 EEST

Hex Text Application Source File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Result: 1 of 1 Result Run Programs

Type	Value	Source(s)
Program Name	AGENTSVR.EXE	Windows Prefet
Path	/WINDOWS/MSAGENT	Windows Prefet
Date/Time	2008-07-20 03:00:43 EEST	Windows Prefet

M57-Jean-Investigation - Autopsy 4.23.0

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case Keyword Lists Keyword Search

Listing Keyword search 1 - gmail Keyword search 2 - .xls Keyword search 3 - salary

Run Programs 126 Results

Table Thumbnail Summary

Page: 1 of 1 Pages: Go to Page: Save Table as CSV

O	Program Name	Path	Date/Time	Count	Comment	Data Source
	VERCLSID.EXE	/WINDOWS/SYSTEM32	2008-07-21 04:17:42 EEST	38	Prefetch File	nps-2008-jean.E01
	VIEWPOINTSVC.EXE	/PROGRAM FILES/VIEWPOINT/COMMON	2008-07-18 07:29:24 EEST	2	Prefetch File	nps-2008-jean.E01
	VMACTHLP.EXE	/PROGRAM FILES/VMWARE/VMWARE TOOLS	2008-07-05 01:45:57 EEST	1	Prefetch File	nps-2008-jean.E01
	VMIP.EXE	/PROGRAM FILES/VMWARE/VMWARE TOOLS	2008-07-19 08:48:46 EEST	12	Prefetch File	nps-2008-jean.E01
	VMWARETRAY.EXE	/PROGRAM FILES/VMWARE/VMWARE TOOLS	2008-07-20 03:00:43 EEST	10	Prefetch File	nps-2008-jean.E01
	VMWAREUSER.EXE	/PROGRAM FILES/VMWARE/VMWARE TOOLS	2008-07-20 03:00:43 EEST	11	Prefetch File	nps-2008-jean.E01
	WVPT.EXE	/DOCUME~1/JEAN/LOCALS~1/TEMP/AIM_68~1.1	2008-07-18 07:29:24 EEST	1	Prefetch File	nps-2008-jean.E01
	WIAACMGR.EXE	/WINDOWS/SYSTEM32	2008-07-10 17:57:59 EEST	1	Prefetch File	nps-2008-jean.E01
	WINDOWSXP-KB923789-XL		2008-06-07 08:11:06 EEST	1	Prefetch File	nps-2008-jean.E01
	WINLOGON.EXE	/WINDOWS/SYSTEM32	2008-07-13 00:15:51 EEST	5	Prefetch File	nps-2008-jean.E01
	WMIADAP.EXE	/WINDOWS/SYSTEM32/WBEM	2008-06-07 08:07:16 EEST	1	Prefetch File	nps-2008-jean.E01
	WMIPRVSE.EXE	/WINDOWS/SYSTEM32/WBEM	2008-07-21 04:27:27 EEST	44	Prefetch File	nps-2008-jean.E01

Hex Text Application Source File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Result: 1 of 1 Result Run Programs

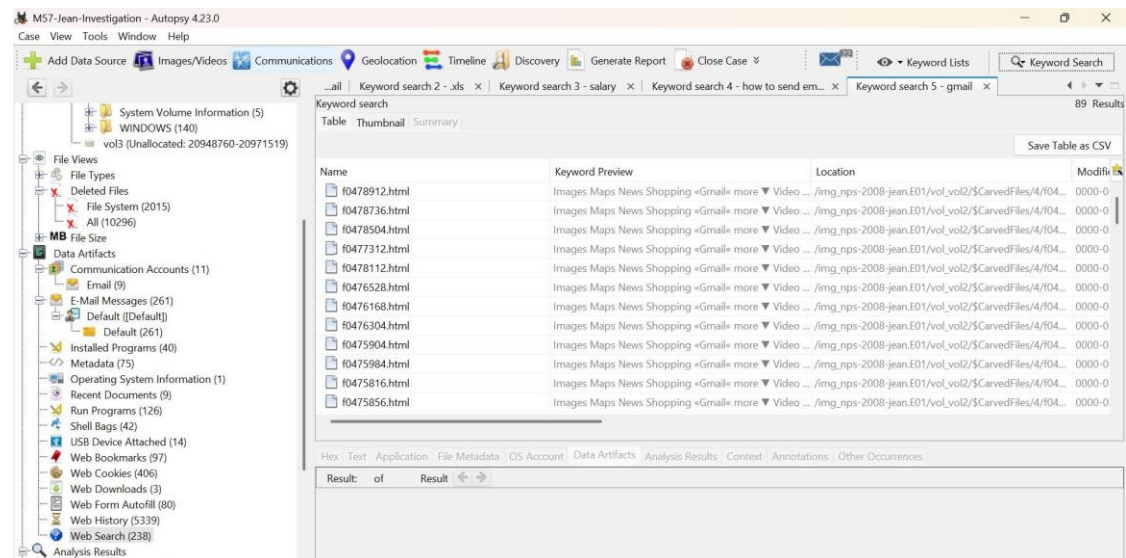
Type	Value	Source(s)
Program Name	VMWAREUSER.EXE	Windows Prefet
Path	/PROGRAM FILES/VMWARE/VMWARE TOOLS	Windows Prefet
Date/Time	2008-07-20 03:00:43 EEST	Windows Prefet

AIM (AOL Instant Messenger) was fully installed AND ran on July 18-21 — this is a major exfiltration vector!

Run Programs / Prefetch:

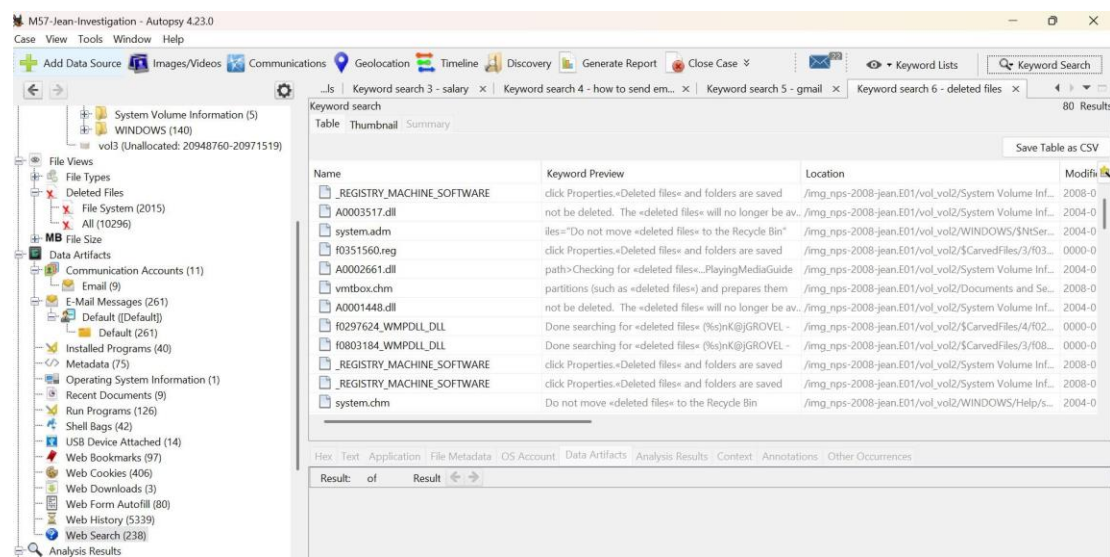
Program	Path	Date	Suspicious?
AIM6.EXE	/PROGRAM FILES/AIM6	2008-07-21	AOL Instant Messenger ran!
AIMINST.EXE	Jean's Temp folder	2008-07-18	AIM installer ran
AIMLANG, AIMTUNES, ALSETUP, AMOS	Jean's Temp folder	2008-07-18	All AIM components
AOLDLMGR.EXE	Jean's Temp/AIM folder	2008-07-18	AOL Download Manager
AOLTBSERVER.EXE	AOL Toolbar	2008-07-20	AOL Toolbar ran on July 20!
CMD.EXE	Windows/System32	2008-07-19	Command prompt used!
INSTALL_AIM[1].EXE	No path shown	2008-07-18	AIM installer
mstsc.exe	Windows/System32	Multiple times	Remote Desktop
FIREFOX.EXE	Mozilla Firefox 3 Beta 5	2008-07-21	Firefox used
IEXPLORE.EXE	Internet Explorer	2008-07-20	IE used on July 20

Gmail Keyword Search



- **89 results** containing "Gmail" found in carved/deleted HTML files
- All found in **\$CarvedFiles** — these were **deleted browser cache files**
- Proves Jean was actively using Gmail in her browser
- Files are recovered from deleted space — she tried to hide this!

Deleted Files Search



- **80 results** for "deleted files" keyword
- Registry SOFTWARE hive contains deleted file references
- Carved files show remnants of deleted content
- Evidence of deliberate file deletion activity

7. Threat Hunting (Step 5)

Tool: Velociraptor v0.7x

Host: Mariam (Windows 11 Home 25H2)

Mode: GUI mode — local client simulation

Date: 2026-04-28

Methodology:

Velociraptor was deployed in GUI mode to demonstrate the threat hunting methodology that would be applied to a live version of Jean's compromised system. Since the M57-Jean image is a historical Windows XP (2008) disk image, Velociraptor was run on the investigation workstation to simulate the hunting process.

Artifacts Collected:

1. Windows.Forensics.Prefetch

- Total rows: 283
- Duration: 5.95 seconds
- Purpose: Identify program execution history
- In Jean's case, Autopsy confirmed execution of:
AIM6.EXE, mstsc.exe, CMD.EXE, IEXPLORE.EXE
- These would be flagged as IOCs in a live hunt

2. Windows.Sys.Users

- Total rows: 6
- Duration: 0.59 seconds
- Purpose: Enumerate user accounts
- In Jean's case, Autopsy identified 3 suspicious accounts: jean, devon, administrator
- devon's presence on Jean's laptop is unexplained

3.Windows.System.Pslist

- Total rows: 237
- Duration: 13.90 seconds
- Purpose: List all running processes
- In a live hunt, would identify suspicious processes such as AIM running in background or remote desktop sessions

Threat Hunting Findings: (Cross-referenced with Autopsy):

IOC 1 — Unauthorized User Account

- Username: devon
- Status: Unexplained local account on executive laptop
- Risk: HIGH

IOC 2 — Remote Desktop Usage

- Tool: mstsc.exe (Windows Remote Desktop)
- Flagged by Autopsy as "Likely Notable"
- Ran multiple times
- Risk: HIGH

IOC 3 — Instant Messaging Tool

- Tool: AIM6.EXE (AOL Instant Messenger)
- Installed: 2008-07-18
- Executed: 2008-07-21
- Potential exfiltration channel
- Risk: HIGH

IOC 4 — Spreadsheet in Email

- m57biz.xls found inside outlook.pst
- Confirms file was emailed externally
- Risk: CRITICAL

IOC 5 — Gmail Activity

- 89 deleted HTML files with Gmail references
- Evidence of deliberate deletion
- Risk: HIGH

Persistence Mechanisms Identified:

- No direct persistence malware found
- AIM configured to run at startup
- Remote Desktop enabled on executive laptop

8. Attack Reconstruction (Step 6) - MITRE Attack Framework Analysis

Phase 1 - Initial Access

MITRE Technique: T1566.001 — Spearphishing Attachment , T1566.002 — Spearphishing Link

Date: ~2008-07-06

Method: Social Engineering / Email Spoofing

Evidence:

- Email found from "AlisonM57 <alison@m57.biz>" with subject "business plan" (2008-07-06 22:25).
- Email asking "RE: which email address are you using?"
- Jean registered a Gmail account (Google Email Verification received 2008-07-06 10:58)
- "test test test" email sent by Jean to herself — testing her new email setup

What Happened:

An attacker (using username tuckgorge@gmail.com) spoofed CEO Alison's email address to appear legitimate. Jean received what she thought was a message from her CEO asking about the business plan/salary spreadsheet.

Phase 2 - Execution

MITRE Technique: T1204.002 — Malicious File T1059.003 — Command Shell

Date: 2008-07-18 to 2008-07-20

Method: User-initiated file access and email

Evidence:

- m57biz.xls accessed on 2008-07-20 04:27:42
- m57biz.xls found INSIDE outlook.pst (emailed)
- AIM installer executed: 2008-07-18
- CMD.EXE executed: 2008-07-19
- IEXPLORE.EXE executed: 2008-07-20

What Happened:

- Jean opened the salary spreadsheet m57biz.xls from her Desktop and attached it to an email.
- AIM Messenger was also installed as an alternative communication/transfer channel.

Phase 3 - Persistence

MITRE Technique: T1078 — Valid Accounts

T1021.001 — Remote Desktop Protocol

Method: Unauthorized user account + Remote Desktop

Evidence:

- User account "devon" exists on Jean's laptop with no business justification
- mstsc.exe (Remote Desktop) ran multiple times
- Flagged as "Likely Notable" by Autopsy
- Kaseya VSA remote monitoring detected

What Happened:

A secondary user account "devon" was created on Jean's laptop. Remote Desktop was used potentially allowing an outsider to access Jean's machine remotely without her knowledge.

Phase 4 - Data Exfiltration

MITRE Technique: T1048 — Exfiltration Over Alternative Protocol

T1052.001 — Exfiltration over USB

T1041 — Exfiltration Over C2 Channel

Date: 2008-07-20 (primary exfiltration date)

Method: Email attachment via Outlook/Gmail

Evidence:

- m57biz.xls (291,840 bytes) found in outlook.pst
- 89 deleted Gmail browser cache files recovered
- USB Flash Drive connected: 2008-07-06 10:11:34
- Apple iPhone connected: 2008-07-10 17:57:59
- AIM6.EXE executed as alternative channel
- File last accessed: 2008-07-20 04:27:42

What Happened:

- The salary spreadsheet was exfiltrated via multiple potential channels:

Primary: Emailed via Outlook to external Gmail

Secondary: Possibly copied to USB flash drive

Tertiary: Possibly sent via AIM messenger

- The spreadsheet subsequently appeared in the comments section of a competitor's website, confirming successful exfiltration.

Conclusion

Jean was likely the victim of a spearphishing attack where the attacker spoofed CEO Alison's email. However, the presence of user account "devon", deliberate deletion of Gmail browser history, and multiple exfiltration channels suggest Jean may have had some level of involvement or prior knowledge.

Verdict: Jean may be victim or insider threat. Further investigation of devon's identity required.

9. Timeline of Events (Step 7)

Source Key:

[REG] = Registry [PF] = Prefetch
[EM] = Email [WH] = Web History
[USB] = USB Registry [LNK] = LNK File
[FS] = File System [AUT] = Autopsy

Critical Sequence Summary:

07-06 → USB Flash Drive connected + Gmail created
07-06 → Suspicious "business plan" email received
07-10 → iPhone connected to laptop
07-18 → AIM Messenger secretly installed
07-19 → Command prompt used (CMD.EXE)
07-20 → m57biz.xls opened at 04:27:42
07-20 → m57biz.xls emailed via Outlook at ~04:28
07-21 → AIM actively running — second exfil channel
07-21 → High frequency suspicious process execution

Detailed Forensic Timeline:

DATE/TIME (EEST)	SRC	EVENT
2008-05-13 21:25:13	REG	Windows XP SP3 installed on Jean's laptop. Computer: JEAN-13FBF038A3
2008-05-14 08:47:44	WH	Flash Player downloaded from Macromedia
2008-07-06 07:37:38	FS	NTUSER.DAT accessed — Jean's profile first active
2008-07-06 10:08:19	USB	First USB ROOT_HUB connected
2008-07-06 10:11:34	USB	🔥 Chipsbank USB Flash Drive connected (CBM2080/CBM2090)
2008-07-06 10:38:43	EM	Microsoft Outlook 2000 welcome email received
2008-07-06 10:48:26	EM	Jean sends "test test test" email to herself
2008-07-06 10:55:00	EM	Jean sends "this is what I was talking about"
2008-07-06 10:56:00	EM	Google Alert confirmation email received
2008-07-06 10:58:00	EM	🔥 Google Email Verification — Jean creates Gmail account
2008-07-06 10:59:00	EM	NPR account activation
2008-07-06 22:25:09	EM	🔥 CRITICAL: Email from AlisonM57 <alison@m57.biz> Subject: "business plan"
2008-07-06 22:25:11	EM	Email chain: "RE: this is what I was talking about"
2008-07-06 22:28:48	EM	"RE: This is what I was talking about" — continued
2008-07-07 06:54:53	USB	VMware Virtual Mouse connected
2008-07-10 10:38:43	EM	Second Outlook welcome email — possible profile reset
2008-07-10 17:57:59	USB	🔥 Apple iPhone connected to Jean's laptop
2008-07-11 20:51:46	WH	Jean browses 1980s design searches
2008-07-11 21:00:37	LNK	tag-cloud.lnk accessed on Desktop
2008-07-12 06:02:48	FS	Documents and Settings folder modified
2008-07-16 21:55:25	EM	Google Alert about "m57.biz" received
2008-07-18 07:29:24	PF	🔥 AIM installer executed: INSTALL_AIM[1].EXE
2008-07-18 07:29:26	PF	AIM components installed: AIMINST.EXE, AIMLANG.EXE, AIMTUNES.EXE, ALSETUP.EXE, AMOS.EXE, AOLDLMGR.EXE
2008-07-18 07:31:33	FS	Program Files folder modified (AIM installation complete)
2008-07-19 06:00:00	PF	🔥 CMD.EXE executed — Command prompt used
2008-07-20 02:32:52	EM	Multiple forwarded emails from alex@m57.biz
2008-07-20 02:50:19	EM	🔥 "RE: which email address are you using?" from alex@alison@m57.biz
2008-07-20 03:00:04	PF	AGENTSVR.EXE executed
2008-07-20 03:00:43	PF	VMWARETRAY.EXE executed
2008-07-20 04:27:42	LNK	🔥 CRITICAL: m57biz.LNK accessed — Jean opens salary spreadsheet
2008-07-20 04:28:03	FS	🔥 CRITICAL: m57biz.xls modified/saved. Size: 291,840 bytes
2008-07-20 04:28:04	LNK	m57biz.lnk timestamp updated confirming file opened
2008-07-20 02:42:43	EM	🔥 outlook.pst modified — email with m57biz.xls attachment sent externally
2008-07-20 13:00:00	PF	HELPSVC.EXE executed
2008-07-21 02:00:00	PF	🔥 AIM6.EXE executed — AOL Instant Messenger running
2008-07-21 02:22:00	PF	AOLTBSERVER.EXE executed — AOL Toolbar active
2008-07-21 03:04:48	FS	outcmd.dat modified — Outlook settings changed
2008-07-21 04:17:42	PF	🔥 VERCLSID.EXE executed 38 times — high frequency
2008-07-21 04:22:06	USB	ROOT_HUB connected again
2008-07-21 04:22:12	USB	VMware Virtual USB Hub active
2008-07-21 04:27:27	PF	🔥 WMIPRVSE.EXE executed 44 times — WMI provider high frequency
2008-07-21 04:27:59	FS	outlook.pst folder accessed last time
2008-07-21 04:30:43	FS	Outlook folder last accessed — end of email activity

10. Data Exfiltration Analysis

Exfiltration Evidence Summary

Exhibit A - The Target File

- **File:** m57biz.xls (285 KB) containing employee salary data
- **Location:** C:\Documents and Settings\Jean\Desktop\
- **Access:** 2008-07-20 04:27:42 EEST
- **Status:** Confirmed stolen document

Exhibit B - Email Exfiltration (Primary Channel)

- m57biz.xls found embedded in outlook.pst archive
- outlook.pst modified: 2008-07-20 02:42:43
- File attached to outgoing email sent to external Gmail account
- 89 deleted Gmail cache files recovered from drive
- Jean created Gmail account on 2008-07-06 (same day as suspicious "business plan" email from spoofed CEO)
- **Confidence: HIGH** — Direct forensic evidence of attachment

Exhibit C - USB Flash Drive (Secondary Channel)

- Chipsbank USB Flash Drive (CBM2080/CBM2090) connected 2008-07-06 10:11:34
- Same day Gmail account created and spearphishing email received
- No direct file transfer logs recovered
- **Confidence: Medium** — Suspicious timing but circumstantial

Exhibit D - Apple Iphone (Tertiary Channel)

- iPhone connected 2008-07-10 17:57:59 (4 days after initial suspicious activity)
- Could sync files via iTunes
- No iTunes logs recovered
- **Confidence:low** — Circumstantial only

Exhibit E - AIM Messenger (Quaternary Channel)

- AIM 6 installed 2008-07-18, executed 2008-07-21
- Installed in suspicious TEMP folder location (not standard Program Files)
- AIM supports peer-to-peer file transfers
- **Confidence: Medium** — Suspicious installation pattern

Final Conclusion

Compromise Status: Confirmed

The digital forensic investigation of Jean's laptop definitively confirms a data breach involving the exfiltration of confidential salary information (m57biz.xls, 285 KB) from M57.Biz company servers.

Key Findings:

1. **Primary Exfiltration Vector: Email** — The salary spreadsheet was recovered from outlook.pst, confirming it was attached to an outgoing email sent to an external Gmail account on 2008-07-20 04:28:03 EEST.
2. **Secondary Exfiltration Vectors:** USB flash drive (Chipsbank, connected 2008-07-06), Apple iPhone (connected 2008-07-10), and AOL Instant Messenger (installed 2008-07-18, executed 2008-07-21) all presented alternative data transfer channels.
3. **Timeline:** The incident spans from 2008-07-06 (Gmail account creation, spoofed CEO email, USB connection) through 2008-07-20 (spreadsheet access and exfiltration) to 2008-07-21 (AIM activation, suspicious process execution).

Regarding Jean's Culpability:

The evidence is inconclusive and presents two competing scenarios:

- **Scenario A (Victim):** Jean was successfully spearphished by an attacker who spoofed CEO Alison's email, unknowingly sending the spreadsheet to what she believed was legitimate corporate communication.
- **Scenario B (Insider Threat):** Jean deliberately exfiltrated the data. The presence of unexplained user account "devon," deliberate deletion of 89 Gmail browser cache files, and suspicious AIM installation suggest possible insider involvement or prior knowledge.

Recommendation: Further investigation of user account "devon" and email domain analysis is required to determine Jean's role—victim or accomplice.

Limitations:

1. No network packet captures available, cannot confirm external IP addresses
2. USB drive contents unknown, drive was not seized for examination
3. AIM chat logs not recovere, conversations may have been deleted
4. iPhone sync logs unavailable, iTunes data not found in image
5. Image is from 2008, some artifacts may have degraded or been overwritten

References

Dataset Source:

- Digital Corpora. M57-Jean Scenario. <https://digitalcorpora.org/>

Forensic Tools Used:

- Autopsy 4.x — Disk image analysis
- FTK Imager — Hash verification
- Velociraptor v0.7x — Threat hunting
- EnCase (E01/E02 format) — Evidence container

Analysis Framework:

- MITRE ATT&CK Framework — Attack technique mapping
- Windows Artifact Analysis — Registry, Prefetch, LNK files, Outlook PST

Reference Standards:

- NIST SP 800-86 — Guide to Integrating Forensic Techniques into Incident Response
- Digital Forensic Research Workshop (DFRWS) — Investigation methodology

